

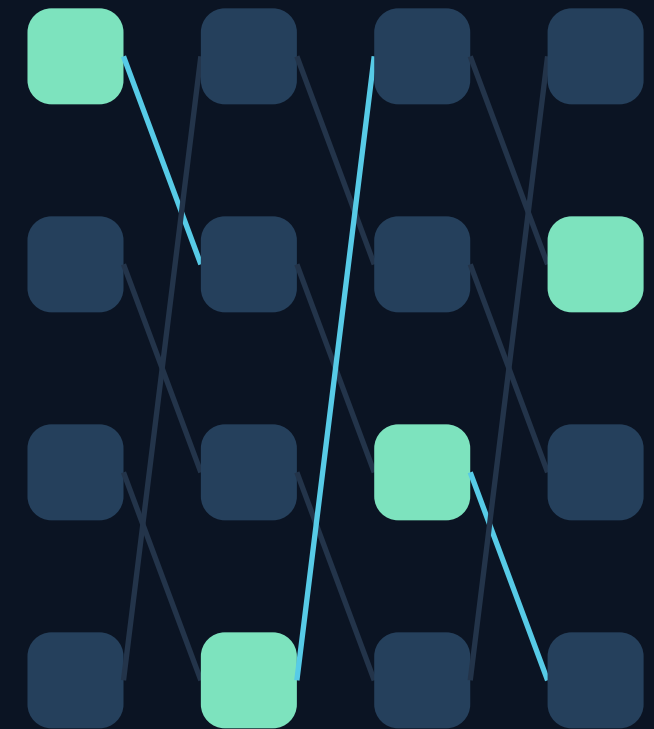
# Sleeping Machines

A trainable substrate for intelligence through time and sparse events.

Build useful capacity faster than active  
computation.  
From datacenters to mobile devices and robotics.

**Tero Keski-Valkama | Sole founder**

Proposed raise €3M · Bullish case €50M pre-money  
Research-stage deep tech · 3 October 2026



**Capacity beyond activity**

# A new model family. A new computing substrate.

The goal: useful intelligence with less active work, persistent state and adaptation.

## CAPACITY BEYOND ACTIVITY

### More memory

Grow available learned state while selecting a small amount of computation for each event.

## TIME COMPUTES

### Learned races

Message content, delays and local state determine which computation happens and when.

## CLOUD TO EMBODIMENT

### One substrate

Pursue models and runtimes first; extend the same principles to edge systems and future clockless silicon.

**The investment thesis is an architecture-runtime-hardware platform, with staged proof.**

# The constraints reach from power grids to robot batteries.

Economic context for efficient intelligence; these figures are not our revenue market.

## DATACENTERS

**485 → 950**

TWh/year, 2025 → 2030 central IEA projection. Power and data movement make system efficiency economically important.

## MOBILE

**5.8B**

Unique mobile subscribers in GSMA's 2026 report. On-device inference creates a prospective route to broad distribution.

## ROBOTICS

**5M**

Industrial robots operating in 2025. Sparse sensing, persistent state and adaptation are prospective use cases.

**Select one measurable initial workload; expand only after quality and system gains hold.**

# Hard sparse computation that can learn its alternatives.

Conceptual event path; the trained language core is implemented in a synchronous software emulator.

## 01 / MESSAGE

### Content + time

Encode incoming signal. Content and persistent memory influence the emission delay.



## 02 / RACE

### Hard winner

Score candidate keys. A temporal race selects an addressed receiver.



## 03 / MEMORY

### Local update

Deliver the selected value and evolve local persistent state through deep layers.



## 04 / LEARNING

### Route credit

Train contents and timings using selected and unrealized alternatives; charge the credit work.

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**Counterfactual credit: learn from selected and unrealized routes**

**Inference selects values sparsely; current training still computes losing alternatives.**

# Three results make the architecture worth scaling.

Completed text8 experiments; all numbers remain exploratory single-seed evidence.

## LANGUAGE QUALITY

# 2.162 bpc

Best p96 / D4 result is slightly ahead of saved one-pass LSTM at 2.171; it uses more work.

## CAUSAL MECHANISM ABLATION

# −0.135 bpc

Adding value-based route credit improves the same p32 / D4 model with +0.30% estimated fitting work.

## CAPACITY / SELECTION

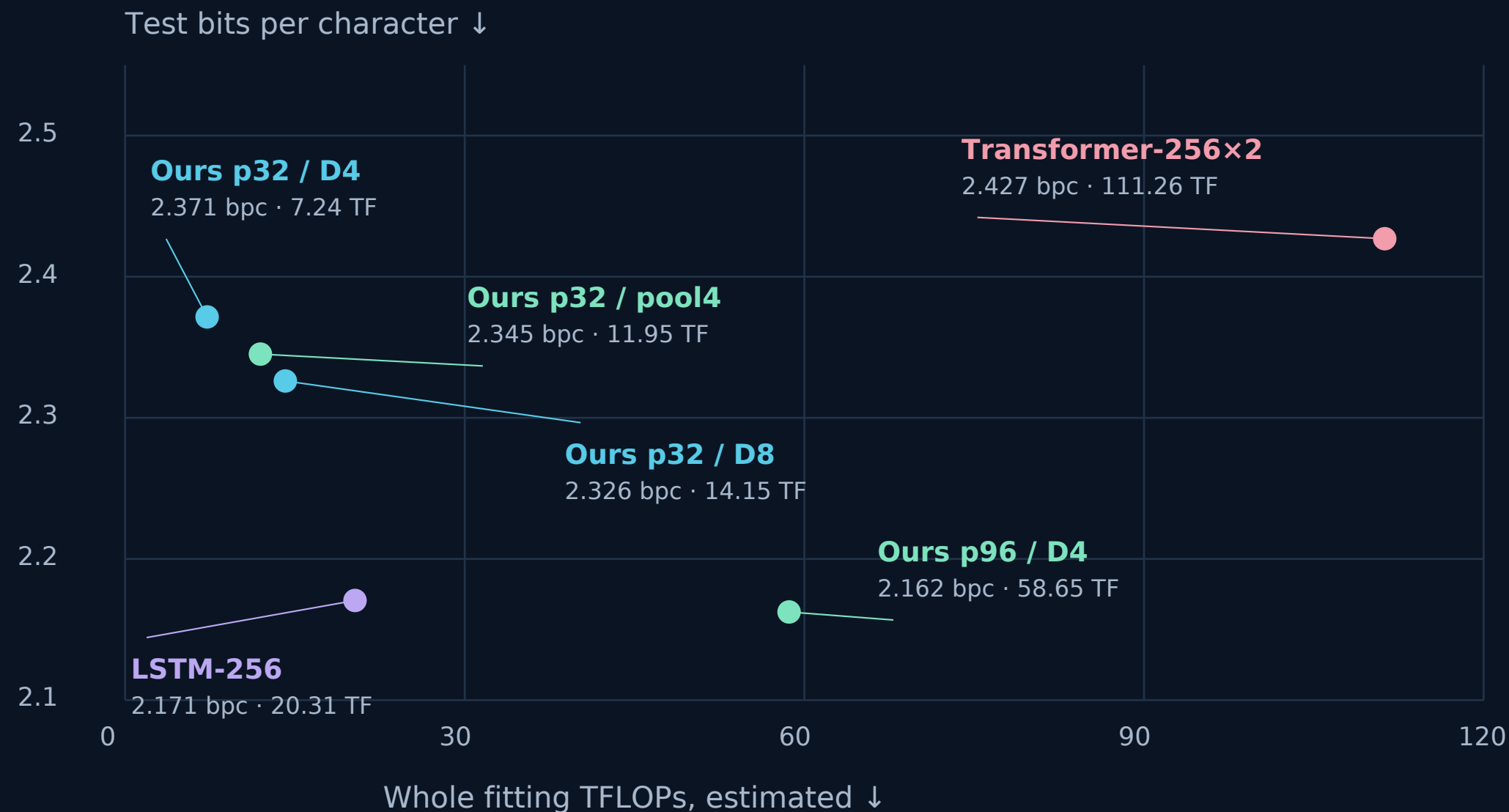
# 8 writes

Doubling the credited p32 memory pool improves quality while keeping selected writes per input position fixed.

**Distinctive mechanisms affect learning; scale, modern controls and serving economics remain open.**

# A promising quality-work frontier at small scale.

Same text8 held-out region and T256 windows; saved one-pass controls, not optimized frontier baselines.



## QUALITY FRONTIER

# 2.162

Ahead of saved LSTM; 2.89× fitting work.

## SMALL TRANSFORMER CONTROL

# 15.4×

p32 / D4: less estimated fitting work and better quality.

BPC ↓ is better. Fitting includes credit/backward/clip/Adam estimates; conventions and training order differ. One seed. No wall-time or energy claim.

# Learning the alternatives changes the result.

The same p32 / D4 model, data budget, seed and evaluation window.

## TEST BITS / CHARACTER

**2.506 → 2.371**

Timing-only credit versus added local alternative-value route credit.

## ESTIMATED FITTING WORK

**+0.30%**

Hard forward selection and parameter count retained.

## CHARACTER PERPLEXITY

**−8.9%**

Derived from the BPC difference; one controlled seed.

## Same p32 / D4 model; hard forward routes retained

Depth-8 also improves by 0.130 bpc with route credit. This supports the credit mechanism across two depths; replication is next.

**A falsifiable architectural mechanism, with a small counted arithmetic increment.**

# More available state. The same eight selected writes.

Matched credited p32 / D4 pool expansion: 16 → 32 slots.

## Pool 2



16 slots · 512 value scalars  
8 writes · 16 keys scored  
2.371 bpc

## Pool 4



32 slots · 1,024 value scalars  
8 writes · 32 keys scored  
2.345 bpc

## WHAT SCALES

# 8 writes

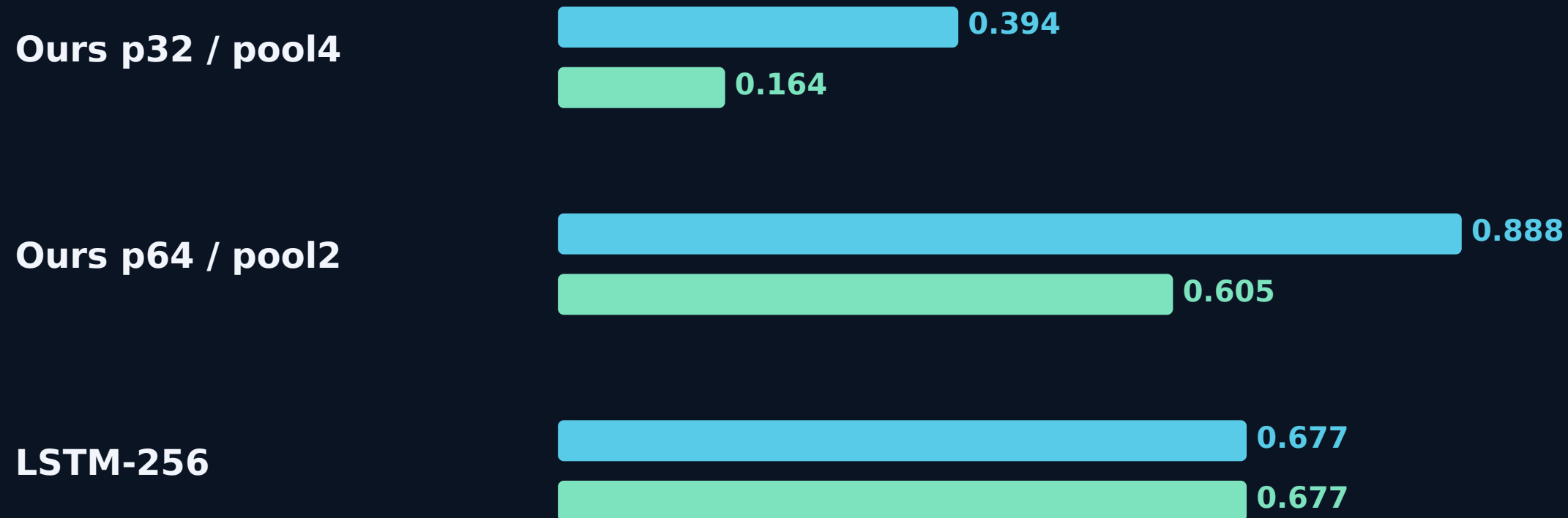
Winner-only arithmetic +0.66%.  
Fitting work +65.0%. Keys and  
value storage double.

**Quality improves by 0.026 bpc; total capacity and learning costs remain explicitly charged.**

# The sparse path is implemented; system proof is next.

Traced arithmetic per input position. Only selected proposal values are computed in the winner path.

Estimated MFLOPs per evaluated input position



## STATUS

All keys remain scored.  
Actual trained FP32 parity and held-out rescore are pending. Traffic, setup, latency and joules must be measured.

Cyan: emulator / control forward Green: winner-only / same control

These are backend/shape estimates. Native evaluation pays overlapping-window warmup (~2 positions/target); LSTM recurrent evaluation differs. Not a service-cost win.

# An adaptation signal exists. Its work is accounted for.

An earlier integrated CPU model, on a new development stream; separate from the latest language fits.

## PREDICT BEFORE UPDATE

**3.191 → 3.096**

8,191 new development targets; same persistent state, frozen versus full-backbone adaptation.

## COSTED LEARNING

**10.7× work**

Total online processing work includes credit, backward and optimizer. Retention under drift and native asynchronous learning remain open.

**Online-learning upside: persistent adaptive systems. Current result: one costed pilot.**

# The contribution is the complete construction.

Established alternatives already address parts of this problem; they remain essential controls.

| Approach             | Established strength                                                          | Our comparison obligation                                               |
|----------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Transformer / MoE    | Strong quality; MoE separates available parameters from selected expert work. | Match quality/data and charge routing, memory traffic and training.     |
| RNN / selective SSM  | Persistent state and efficient sequence processing.                           | Test whether temporal races add useful quality or allocation.           |
| Neuromorphic systems | Event-driven hardware and local learning precedents.                          | Prove expressive model quality, mapping and measured energy.            |
| Sleeping Machines    | Joint timing, sparse memory, deep representation and alternative credit.      | Scale integrated mechanisms without replacing them with dense carriers. |

**Sparsity, recurrence and on-chip learning alone are not the novelty claim.**

# Enter through software. Earn the hardware expansion.

Proposed commercialization sequence; no customers or partner interest yet.

## 01 / MODEL + RUNTIME

### First wedge

License a validated model/runtime for a chosen persistent sequence workload. Demonstrate quality and complete serving cost.

## 02 / ACCELERATOR CO-DESIGN

### Partner path

Port verified kernels, profile memory and event execution, then pursue a joint platform or IP license.

## 03 / CLOCKLESS SUBSTRATE

### Long-term upside

Explore delay precision, sparse local state and event scheduling in FPGA/ASIC feasibility work after software gates pass.

**The €3M round funds evidence and feasibility; a full advanced-node tapeout is outside this budget.**

# Incumbents have reasons to evaluate a new substrate.

Potential counterparties only. Strategic relevance is an inference; no relationship is claimed.

## AMD

### Models + silicon

Silo AI acquisition and Liquid AI collaboration show appetite for model/software capabilities. Earn attention with AMD-compatible scale and measured gains.

## INTEL

### Event systems

Hala Point and Loihi 2 establish event-computing research. Offer a credible expressive model and hardware mapping experiment.

## GOOGLE

### Full-stack co-design

TPU 8t/8i embody model-hardware optimization. Offer independently validated improvements that complement a large existing research stack.

**A measured advantage could become strategic leverage against incumbent economics; interest must be established.**

# AMD's Silo AI deal supports the strategic logic.

Announced approximately \$665M all-cash acquisition in 2024; a mature-company precedent.

## WHAT AMD ACQUIRED

### Team + stack

AI scientists and engineers, enterprise solutions, model development and software capability on AMD hardware.

## THE LUMI CONNECTION

### Pororo / Viking

AMD highlighted multilingual models and training on AMD-powered LUMI. LUMI is the supercomputer, not the model name.

## OUR LESSON

### Prove leverage

Build a capability that improves a platform's economics and adoption. One small research project is not a direct pricing comparable.

**Efficient models can matter to a chip ecosystem; training on its chips is one part of the value case.**

# Capture a fraction of verified value created.

Illustrative annual economics, contingent on quality and complete system savings.

| Scenario input     | Arithmetic                                     | Potential annual revenue |
|--------------------|------------------------------------------------|--------------------------|
| Datacenter license | €1B eligible spend × 20% savings × 20% capture | €40M                     |
| Mobile deployment  | 100M licensed new devices × €0.50 royalty      | €50M                     |

Datacenter customer retains €160M of the assumed €200M gross savings before migration and other costs. Robot fleets could support per-system adaptation licenses after validation.

**These are assumptions for value-capture design, not booked revenue, forecasts or an additive TAM.**

# A founder at the intersection of models and systems.

Public professional record and a primary published invention document.

Tero  
Keski-Valkama

Sole founder

Public work spans machine learning, software architecture and industrial applications. The relevant founder case is sustained technical execution across these layers.

2012-2018

Cybercom · Software Architect

Historical role in public professional profile.

2018-2022

HERE · Lead ML & AI Engineer

Self-reported research and applied ML leadership.

2022-2025

Alloy.ai → Kaiko.ai

Staff engineering; multimodal foundation-model work.

2023 publication

EP4148389A2 · Named inventor

Maps from geospatial observations; applicant HERE.

Career dates are self-reported. No PhD, current employer or ownership of HERE inventions is asserted. Sole-founder status supplied by Tero.

# Build a moat around evidence, execution and integration.

The repository is now private; future progress disclosure needs a deliberate policy.

## RESEARCH + DATA ROOM

### Auditable

Versioned results, ablations, numerical contracts and retained negative findings support credible technical diligence.

## RUNTIME + CO-DESIGN

### Compound

Model, credit estimator, sparse execution, measurement and hardware mapping can become mutually reinforcing know-how.

## IP + ADOPTION

### To establish

Clarify contribution rights, licensing and employer assignments; then build deployment tooling and partner-specific integration.

**Private source control helps disclosure discipline; exclusive IP and adoption still require proof.**

# Use capital to retire the decisive uncertainties.

Milestones are proposed gates, not completed deliverables or guaranteed dates.

## MONTHS 0-3

### Reproduce

Hire initial team. Admit trained sparse parity, complete 90M evidence and three-seed comparisons. Resolve IP and data-room rights.

## MONTHS 3-6

### Measure

Benchmark latency, traffic, memory and joules on one chosen workload with tuned controls. Target  $\geq 20\%$  full-system cost reduction at matched quality.

## MONTHS 6-12

### Demonstrate

Pursue independent reproduction and prospective design-partner trials. Validate willingness to pay and a deployable runtime.

## MONTHS 12-18

### Expand

Scale a winning workload; test drift/retention and a hardware feasibility path. Raise the next round only on stronger evidence.

**If a gate fails, diagnose or narrow the wedge before increasing compute or committing to silicon.**

# €3M to scale research and engineering aggressively.

Proposed 18-month budget; assumptions to refine with hiring and compute quotes.



- Team: research + systems **€1.35M | 45.0%**
- Compute + independent replication **€0.90M | 30.0%**
- Hardware feasibility + measurement **€0.25M | 8.3%**
- IP, legal, operations **€0.20M | 6.7%**
- Contingency reserve **€0.30M | 10.0%**

**€2.70M operating plan + €0.30M reserve · Average planned operating burn €150k/month.**  
Team assumption: average six FTE including founder at €150k/year fully loaded. No full advanced-node tapeout allocation.

# Price the platform option with explicit assumptions.

€50M is a bullish negotiating case; €100M is a stretch case requiring substantially stronger conviction.

## BULLISH PRICED ROUND

**€50M**

€3M raise → €53M post-money;  
5.66% new-investor ownership.

## REVERSE UNDERWRITING

**6.74%**

Required platform-success  
probability in the stated €10B exit  
scenario.

## STRETCH SCENARIO

**€100M**

€100M pre-money requires 13.49%  
in the same model; 2.91%  
ownership.

$$V = p \times \text{exit equity} \times \text{future retention} / (1 + \text{discount rate})^{\text{years}}$$

Assumptions: €10B exit; 30% stake retention; 10 years; 15% discount; failure value zero.

**Required odds are reverse-calculated conditions. The benchmarks do not estimate those probabilities.**

# Successful platform adoption can support very large outcomes.

Conditional commercial arithmetic illustrates the scale required; no forecast or market multiple claim.

| Successful outcome assumption    | Value capture assumption                                       | Equity-value scenario |
|----------------------------------|----------------------------------------------------------------|-----------------------|
| €10B/year eligible compute spend | 20% savings × 20% capture = €400M/year; 25× revenue assumption | €10B                  |
| €50B/year eligible compute spend | 20% savings × 25% capture = €2.5B/year; 20× revenue assumption | €50B                  |

A cross-platform substrate could participate in cloud inference, persistent agents and embodied systems. Reaching these outcomes requires sustained quality, scale, margins and broad adoption.

**The upside is large if the platform works and wins adoption. Its probability remains unmeasured.**

# Fund the evidence that makes a new substrate investable.

Private discussion draft · €50M pre-money bullish case · No customer interest reported.

# €3 million

Scale the integrated architecture.  
Prove the economics. Build the platform.

**01**

Replicate quality and counterfactual learning at meaningful scale.

**02**

Demonstrate sparse serving gains with complete resource boundaries.

**03**

Turn verified advantages into partnerships and a hardware roadmap.

Tero Keski-Valkama · Private investor diligence available by arrangement

# Quality and work, in the same units for every model.

Completed one-pass 10M text8 fits; T256 held-out evaluation; estimated operation counts.

| Model                   | Test bpc | Params K | Fit total TF | Fit MF/pos | Emul MF/pos | Winner MF/pos |
|-------------------------|----------|----------|--------------|------------|-------------|---------------|
| p32 / D4 timing-only    | 2.506    | 108.9    | 7.22         | 0.722      | 0.240       | 0.163         |
| p32 / D4 credited       | 2.371    | 108.9    | 7.24         | 0.725      | 0.240       | 0.163         |
| p32 / D4 pool4 credited | 2.345    | 177.0    | 11.95        | 1.196      | 0.394       | 0.164         |
| p32 / D8 credited       | 2.326    | 210.0    | 14.15        | 1.416      | 0.463       | 0.310         |
| p64 / D4 credited       | 2.183    | 422.5    | 26.79        | 2.680      | 0.888       | 0.605         |
| p64 / D4 pool4 credited | 2.179    | 689.8    | 44.07        | 4.410      | 1.458       | 0.607         |
| p96 / D4 credited       | 2.162    | 940.9    | 58.65        | 5.868      | 1.947       | 1.324         |
| LSTM-256                | 2.171    | 338.4    | 20.31        | 2.032      | 0.677       | 0.677         |
| Transformer-256x2       | 2.427    | 1658.9   | 111.26       | 11.133     | 3.710       | 3.710         |

Fit MF/pos = per training position. Inference MF/pos = per evaluated input position. Native quality uses training evaluator; winner-only trained parity/rescore pending. Native overlap warmup adds work per scored target. Full 15-row CSV retains older models.

# The comparison boundaries are part of the evidence.

Maintain the original records and disclose what a result actually measures.

| Boundary          | Current completed evidence                                                                      | Remaining work                                                                |
|-------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Data / evaluation | Native: text8[0:10M], random 64×128 segments, ~9.994M positions; test[95M:96M], T256.           | Tail target counts and training order differ from E64; align modern controls. |
| Learning / seeds  | 1,220 native updates (latest rows), lr .004 cosine, one seed; state reset per fit segment.      | Replication, robust recipe, depth/horizon and 90M completed held-out results. |
| Sparse execution  | All keys scored; selected values only in winner evaluator; small random FP64 equality fixtures. | Actual trained FP32 winner/state/cache equality and held-out rescore.         |
| System economics  | Arithmetic traces/formulas; persistent local state within windows.                              | Setup, warmup, traffic, memory, throughput, latency, joules and total cost.   |

# Keep the failures beside the positive mechanism results.

A disciplined research program should revise interpretations when evidence changes.

| Finding                                      | Interpretation                                                                     | Response                                                                 |
|----------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Uncredited D8 pool expansion worsens quality | More capacity needs learnable selection; it is not automatically useful.           | Retain negative pair; test credited expansion with matched depth/seed.   |
| Write-credit variants regress or diverge     | A particular local estimator is unstable; not all route credit is equivalent.      | Withdraw variants; preserve original result and derive corrected credit. |
| Best p96 costs more than LSTM                | Small quality lead is not comparable-quality resource superiority.                 | Tune and scale; require complete quality-cost frontier.                  |
| Event-vision screening is mixed              | Small coarsening gains do not clear every full-screen seed or strong RBF controls. | Avoid cross-domain supremacy claim; keep modality-specific gates.        |

# The price depends on dilution, exit scale and time.

Required platform-success odds at 10 years and 15% discount; zero failure value.

| Exit equity | Stake retention | Required p: €50M | Required p: €100M |
|-------------|-----------------|------------------|-------------------|
| €3B         | 10%             | 67.43%           | 134.85%           |
| €3B         | 30%             | 22.48%           | 44.95%            |
| €3B         | 50%             | 13.49%           | 26.97%            |
| €10B        | 10%             | 20.23%           | 40.46%            |
| €10B        | 30%             | 6.74%            | 13.49%            |
| €10B        | 50%             | 4.05%            | 8.09%             |
| €30B        | 10%             | 6.74%            | 13.49%            |
| €30B        | 30%             | 2.25%            | 4.50%             |
| €30B        | 50%             | 1.35%            | 2.70%             |

Sensitivity analysis, not observed probabilities. No preferred-stock waterfall, intermediate distributions, follow-on investment cash flows or option-pool adjustment modeled.

# A private evidence pack and primary external context.

Source labels in slide footers are clickable; companion files must travel with the PDF.

**R1 Private completed language evidence ledger (3 October 2026)**

Saved native fits and E64 controls. Original JSON records and SHA256 hashes are supplied in the private diligence pack.

**R2 Private online-learning experiment, 8,191 targets**

Predict-before-update pilot: quality and complete processing-operation estimates; one checkpoint, stream and learning rate.

**R3 Architecture, numerical contracts and report**

Private technical report. Temporal races, counterfactual credit and native sparse inference; trained-weight sparse parity remains pending.

**M1 IEA — Key Questions on Energy and AI, executive summary**

485 TWh datacenter electricity in 2025; central projection 950 TWh in 2030. Sector context, not addressable revenue.

**M2 GSMA — The Mobile Economy 2026**

5.8 billion unique mobile subscribers and 8.8 billion connections. Subscribers are not annual device sales or customers.

**M3 IFR — World Robotics 2026 release, 24 September 2026**

5 million operating industrial robots in 2025 and more than 600,000 annual installations; no adoption by this project.

**T2 Fedus et al. — Switch Transformers**

Capacity beyond selected computation has established MoE precedent; sparsity alone is not a novelty claim.

# Public precedents establish relevance, not endorsement.

Historical roles are self-reported; inventorship is checked against the primary published document.

**S1**      **AMD — Silo AI acquisition completion, 12 August 2024**

Announced approximately \$665M all-cash acquisition; Poro and Viking on AMD. Mature-team strategic precedent, not a pre-seed valuation comparable.

**S2**      **AMD — Acquisition announcement and LUMI collaboration, 10 July 2024**

AMD highlighted enterprise solutions, expertise, software and Poro/Viking training on LUMI using AMD hardware; no single exclusive acquisition cause established.

**S3**      **Liquid AI — \$250M Series A announcement, 13 December 2024**

Financing announcement and AMD collaboration on efficient models; funding amount is not a disclosed company valuation.

**S4**      **AMD Ventures — investment focus**

AI models, infrastructure and silicon investment focus; published Series A-C focus means this project may be earlier than typical admission.

**S5**      **Intel — Hala Point research system, 17 April 2024**

Loihi 2 neuromorphic research prototype, event computing and prospective continuous learning. No Intel interest in this project claimed.

**S6**      **Google — Eighth-generation TPU architecture announcement**

TPU 8t/8i, model-hardware co-design and sparse MoE infrastructure. Existing internal capability creates both potential fit and competition.

**F2**      **European Patent Office — published application EP4148389A2**

Primary published document names Tero Juhani Keski-Valkama as inventor and HERE Global B.V. as applicant. Experience evidence, not Sleeping Machines-owned IP or proof of a grant.